Tony Li

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Education

08/2024–Present **Stony Brook University**
Ph.D. in Computer Science; Advisor: Xiaojun Bi

08/2020–06/2024 **University of Science and Technology of China (USTC)**
B.E. in Computer Science and Technology

Experience

11/2026–05/2027 **Student Researcher, Google**, Mountain View, CA

- Research on **personalized natural language generation** for AI writing assistants, modeling users' writing styles and intended content.

06/2026–08/2026 **GenAI Research Intern, Samsung Ads**, Mountain View, CA

- Developed **cacheable, target-aware user representations** for advertising recommendation under a **256-byte per-user cache budget**.
- Investigated **multi-slot user modeling**, **LLM-based representation learning**, and **knowledge distillation** for efficient personalization.

08/2024–Present **Graduate Research Assistant, Stony Brook University**

- Developed **LLM-based keyboard decoders** for personalized vocabulary adaptation (LEXGR) and ten-finger touchscreen typing (KeySense, CHI 2026).
- Co-led a **decoder analysis framework** (DecoLego) and investigated gesture/tap input across mobile, AR, and VR interfaces.

07/2023–04/2024 **Research Intern, Simon Fraser University**; Advisor: Xing-Dong Yang

- Developed a **battery-free wireless textile sensing interface** and computational tools for textile haptics (CHI 2025).

Selected Publications

Under Review **Li, Tony**, Chen, Z., Zhu, P., Ma, Y., and Bi, X. LEXGR: Extending Flexible LLM-Based Keyboard Decoding to New Words

Under Review Chen, Z.*, **Li, Tony***, Zhu, P., and Bi, X. DecoLego: A Functional Framework for Probing, Recomposing, and Extending Keyboard Decoders

Under Review Newton, F., **Li, Tony**, Bi, X., and Leiva, L. A. Decoding without Profiling: Privacy-Preserving Gesture Typing through Backtracking

Under Review Chen, Z., Zhu, P., Chen, Y., **Li, Tony**, Gu, X. D., Miao, X., Yu, L., and Bi, X. Structure Before Prose: A Persistent Argument Representation for Source-Grounded Legal Writing

CHI'26 **Li, Tony**, Ma, Y., Li, Z., Yu, C., Ramakrishnan, I. V., and Bi, X. KeySense: LLM-Powered Hands-Down, Ten-Finger Typing on Commodity Touchscreens

CHI'25 Xu, W., **Li, Tony**, Wang, Y., Yang, X.-D., and Wu, T.-Y. BIT: Battery-free, IC-less and Wireless Smart Textile Interface and Sensing System

VRW'25 Ma, Y., **Li, Tony**, Li, Z., and Bi, X. LLM-powered Text Entry in Virtual Reality

Technical Skills

Programming	Python, C/C++, C#, Java, SQL, Bash
ML Frameworks	PyTorch, TensorFlow, Hugging Face Transformers
LLM / GenAI	Supervised fine-tuning (SFT), PEFT/LoRA, knowledge distillation, reinforcement learning (RL), RLHF, preference optimization (DPO)
User Modeling	Representation learning, personalization, recommendation/ranking systems, target-aware modeling, sequence modeling
Development	Unity, Android Studio, Flask, Django; AR/VR prototyping